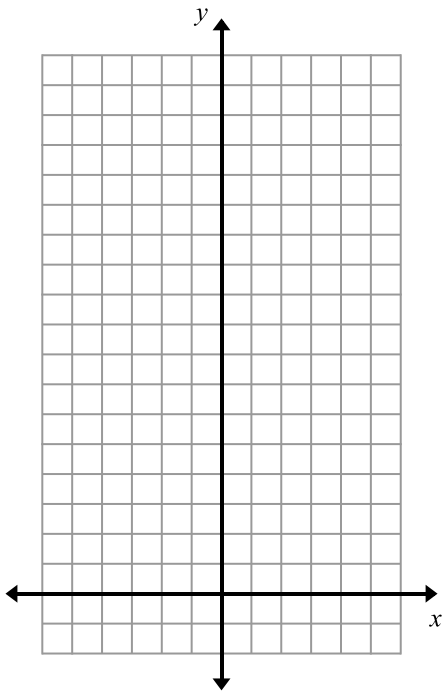
1. Suppose $\vec{r}(t) = \left(A\cos(kt)\right)\vec{i} + \left(A\sin(kt)\right)\vec{j}$, where A and k are positive constants, gives the position of a particle at time t. Find the particle's velocity vector and acceleration vector. Then find the particle's speed when $t = \frac{\pi}{3k}$. Please simplify your final results completely.

2. If $\vec{a}(t) = -9.8\vec{j}$, $\vec{v}(0) = \left(v_0\cos(\alpha)\right)\vec{i} + \left(v_0\sin(\alpha)\right)\vec{j}$ and $\vec{r}(0) = x_0\vec{i} + y_0\vec{j}$, then find the position function $\vec{r}(t)$

3. A particle moves on the plane so that its position at time t is given by the position vector $\vec{r}(t) = \langle t, t^2 \rangle$. Find both $\vec{v}(t)$ and $\vec{a}(t)$. Sketch the graph of the path traced out by the particle along with $\vec{v}(2)$ and $\vec{a}(2)$ drawn as vectors with their tails at the point (2, 4).



4. Suppose that a projectile is fired over level ground with an initial speed of 500 m/sec at an angle of elevation of 60°.

   a. How long will it take before the projectile hits the ground and how far away from the spot from which it was fired will it be at this time?

   b. How high overhead will the projectile be when it is 5 km away from the spot from which it was fired and what is its speed at this time?

   c. What is the maximum height attained by the projectile?

5. A spring gun at ground level fires a golf ball at an angle of 45°. The ball lands 10 m away. What was the ball's initial speed? For that same initial speed, what are the two firing angles that will cause the ball to land 6 m away?

6. A golf ball is hit from the tee to an elevated green that is 45 ft above the tee. To reach this green, the ball must travel over a ravine that ends 360 feet away from the tee. Will the ball clear this ravine and land on the green if the ball was hit with an initial speed of 116 ft/sec. at an angle of elevation of 50°?

7. A human cannonball is to be fired with an initial speed of 85 ft/sec. The person performing this stunt, who must of course be of the right caliber, wishes to land softly on a special cushion located 200 ft away from the spot from which the cannon will be fired. The circus is being held in a large arena with a flat ceiling 75 feet high. Is it possible to fire the cannon so that the performer lands on the cushion without striking the ceiling? If so, at what angle(s) of elevation can the cannon be fired to accomplish this daring feat?